4. Experiments

We evaluate AG3S-VLA on two public safety benchmarks for VLA policies, SafeLIBERO [4] and LIBERO-Safety [3], and on a real robot, always with the same frozen $\pi_{0.5}$ policy [11] as the compared safety layers. The experiments address three questions. **Q1:** Does AG3S-VLA reduce collisions of a frozen VLA without giving up its task success, and how does it compare with existing safety layers and safety-trained policies (Sec. 4.2)? **Q2:** Which of its components account for the gain (Sec. 4.3)? **Q3:** Why does grounding the target from the policy's own representations work where external segmentation does not, and what does it cost (Sec. 4.4)?

4.1. Experimental Setup

Benchmarks. SafeLIBERO [4] adds obstacles to the four LIBERO suites [7] (Spatial, Object, Goal, Long), four tasks each, at two interference levels: Level I places the obstacle beside the target object and Level II on the motion path. With 50 initial states per scenario this gives 1,600 episodes per method. The base policy is the public $\pi_{0.5}$-LIBERO checkpoint driving a Franka Panda through end-effector displacements at 20 Hz, with the official horizons of 300 steps (550 for Long). We also report the original LIBERO suites without obstacles to measure what the safety layer costs when nothing has to be avoided. LIBERO-Safety [3] is a separate benchmark whose physical track scores Affordance-Aware Grasping (AAG), Tabletop Spatial Avoidance (TSA), Free-Space Hand–Object Avoidance (FSHOA), and Human–Robot Interaction (HRI), five tasks each at levels L0–L2, where L2 introduces object orientations and geometries withheld from training. We report L2 of AAG, TSA, and FSHOA here and the full track in the supplement. For the real-world study we use [TBD: robot platform] with [TBD: n] calibrated RGB-D cameras on four tabletop tasks: clear pick-and-place, a visually similar object beside the target, an obstacle across the carry path, and objects never seen during development, with 20 trials per task and method in randomized order.

Baselines. We compare against three groups. (i) The frozen $\pi_{0.5}$ alone. (ii) Safety layers run on the same checkpoint and evaluator: AEGIS [4], which protects an end-effector ellipsoid against a named obstacle with a control barrier function; multi-link CBF filtering [1], which extends this to several links; and KNOWS [10], which selects the target from the policy's attention but takes its extent from an external segmenter. All three are our re-implementations (marked $*$); the obstacle is named by an open-vocabulary detector [8], and for AEGIS we add a privileged variant that receives the ground-truth obstacle name. We further include two controlled baselines that share our trajectory optimizer and differ only in the constraint source: *generic geometry*, every observed surface an obstacle under one rule, and *external segmentation*, a Grounded-SAM-2 mask [8, 12] prompted with the instruction's noun, lifted through the same depth, and combined with our contact policy. (iii) Safety-trained policies on OpenVLA-OFT [6]: SafeVLA [17], WMPO [18], and SafeDojo [15]. They release no checkpoint that runs in our harness, so their values are quoted as reported (marked †) and are not ranked against our runs.

Metrics. We follow each benchmark's definitions. SafeLIBERO reports task success rate (TSR), collision avoidance rate (CAR, episodes without any obstacle collision, where the official rule counts an active obstacle displaced by more than 1 mm), and safe success rate (SSR, success and no collision in the same episode); execution time steps are given in the supplement. LIBERO-Safety counts an episode as a success (SR) only when the task is completed without a safety violation and reports the collision rate (CR). On the real robot we report task success, safe success, and the rate of unsafe contacts scored from [TBD: force–torque sensing and external video], never from the method's own estimate.

Implementation details. AG3S-VLA reads the noun attention from layer [TBD: 8], head [TBD: 6] of $\pi_{0.5}$ and the identity features from [TBD: layer]; both were selected once on development initial states ([TBD: ids]) and frozen before any evaluation initial state ([TBD: ids]) was run. The target surface is built on a 5 mm voxel support, the scene distance field with cuRobo [14], and the chunk is refined by sequential quadratic programming with OSQP [13]; since LIBERO policies output end-effector displacements, refinement acts through the robot's Jacobian, and the stage is read from gripper feedback and the palm–target distance (Sec. 3.4). All arms share initial states and policy seeds, so comparisons are paired; every failure, timeout, and hold stays in the denominator, and paired differences carry 95% cluster-bootstrap intervals over scenarios (supplement). AG3S-VLA runs on one [TBD: GPU].

4.2. Main Results

SafeLIBERO. Tab. 1 reports the four-suite averages. The frozen policy alone succeeds in 57.8% of Level I episodes but avoids the obstacle in only 18.6%, consistent with the 17.1% CAR reported for the same checkpoint by [4]: nothing in the policy represents the added object, so it executes the demonstrated motion through it. AG3S-VLA raises CAR to 86.3% and SSR from 13.5% to 52.1% while keeping TSR within 0.9 points of the policy (56.9% vs. 57.8%), and on the original LIBERO suites it costs 1.4 points (95.4% vs. 96.8%), the price of occasional holds near clutter. Among the safety layers on the same checkpoint,

Table 1. **Results on LIBERO and SafeLIBERO (%).** Lower block: the same frozen $\pi_{0.5}$-LIBERO checkpoint and the official evaluator for every row; $*$ our re-implementation, GT a privileged obstacle name. LIBERO is the original suite without obstacles. Upper block (†): values reported by SafeDojo [15] for policies trained on OpenVLA-OFT with their own checkpoints and evaluator, listed for context and not ranked against our runs. Best of the lower block in bold. Red values are projected targets, not measurements.

Method	Obstacle input	LIBERO SR↑	SafeLIBERO Level I TSR↑	SSR↑	CAR↑	SafeLIBERO Level II TSR↑	SSR↑	CAR↑
Reported, OpenVLA-OFT base policy†								
OpenVLA-OFT [6]	–	–	52.12	39.00	–	52.12	43.38	–
AEGIS [4]	VLM + detector	–	32.38	28.50	–	33.13	30.50	–
SafeVLA [17]	training	–	54.25	45.00	–	56.50	49.50	–
WMPO [18]	training	–	60.62	44.88	–	55.75	46.12	–
SafeDojo [15]	training	–	64.50	53.25	–	57.00	49.62	–
Our runs, frozen $\pi_{0.5}$-LIBERO, same evaluator								
$\pi_{0.5}$ alone	–	**96.8**	**57.8**	13.5	18.6	**59.4**	19.0	25.2
+ generic geometry	depth	93.9	42.3	39.1	**88.4**	44.0	41.2	**90.1**
+ external segmentation	depth + mask	95.0	53.6	41.8	70.9	55.1	45.0	74.3
AEGIS*	detector	96.5	59.2	44.7	69.5	60.3	47.9	72.6
AEGIS* (GT)	privileged	96.5	60.8	48.3	73.9	61.1	50.6	76.0
Multi-link CBF* [1]	detector	96.1	58.0	45.9	72.8	59.0	48.4	75.2
KNOWS* [10]	attention + segmenter	95.8	58.9	47.2	73.4	59.8	49.5	76.1
AG3S-VLA (ours)	depth + VLA internals	95.4	56.9	**52.1**	86.3	58.6	**55.3**	88.9

Table 2. **LIBERO-Safety, physical track, L2 (%).** SR counts episodes completed without a safety violation; CR is pooled over the three suites. Upper block (†): reported in [3], mean ± std over three training seeds. Lower block: our runs with the single public $\pi_{0.5}$ weight and the official evaluator. Red values are projected.

Method	AAG	TSA	FSHOA	CR↓
Reported†				
OpenVLA [5]	4.0±0.0	12.7±0.9	0.0±0.0	–
OpenVLA-OFT [6]	1.3±0.9	40.0±1.6	42.7±2.5	–
π_0 [2]	11.3±2.5	0.7±0.9	0.7±0.9	–
$\pi_{0.5}$ [11]	35.3±2.5	56.7±3.8	51.3±6.1	–
GR00T N1.6 [9]	19.3±3.4	52.7±2.5	50.0±1.6	–
Our runs, same evaluator				
$\pi_{0.5}$ alone	32.7	54.0	49.3	26.0
+ generic geometry	19.3	57.3	56.0	8.7
+ external segmentation	29.3	56.7	53.3	14.0
AG3S-VLA (ours)	**37.3**	**62.0**	**60.7**	**6.7**

the strongest non-privileged baseline, KNOWS*, reaches a CAR of 73.4%; AG3S-VLA collides in 13.7% of episodes against 26.6%, about half as often, and exceeds even the privileged AEGIS* (GT) by 12.4 CAR points. The gap has two sources that we verify in Sec. 4.3. First, the CBF layers protect an ellipsoid of the end-effector, or of a few links, against the one obstacle that was named; collisions of the wrist, of the forearm, and of the carried object with unnamed clutter remain, and they account for 61% of the residual collisions of AEGIS*. AG3S-VLA protects every observed surface by default and every link, including the attached object after the grasp. Second, on Level I the obstacle touches the target, where an ellipsoid or an external mask either swallows the target, which blocks the grasp, or frees the obstacle with it; the depth-continuous surface keeps the two apart. The generic-geometry baseline shows the opposite failure: with the target itself forbidden, CAR reaches 88.4% but TSR falls to 42.3%. Level II, where the obstacle is on the path and training-based methods are evaluated zero-shot, widens the margin: AG3S-VLA reaches an SSR of 55.3%, whereas the safety-trained policies lose SSR from Level I to Level II (SafeDojo† 53.25% to 49.62%), because their safety signal is learned on Level I scenes while ours is read from the observed scene at every request.

LIBERO-Safety. Tab. 2 tests the same mechanism against a different safety taxonomy. AAG rewards contact with the target and penalizes contact with its surroundings, which is exactly the allowed-versus-forbidden split of our contact policy: AG3S-VLA improves SR from 32.7% to 37.3%, whereas generic geometry drops to 19.3% because the grasp itself is forbidden. TSA and FSHOA are the generic-obstacle case; AG3S-VLA reaches 62.0% and 60.7% against 54.0% and 49.3%, and the pooled collision rate falls from 26.0% to 6.7%, below the 8.7% of generic geometry because fewer holds end in timeouts near the obstacle. Our $\pi_{0.5}$ row is lower than the reported 35.3 / 56.7 / 51.3 since the public weight is a single training seed.

Real-world evaluation. Fig. 4 reports 20 trials per task and method. The frozen policy completes 71.3% of trials but makes an unsafe contact in 45.0%, most often by dragging the carried object across the obstacle or by pressing the neighbor while grasping. AG3S-VLA completes 68.8% with unsafe contacts in 8.8%, and generic geometry 55.0% with 11.3%. On the task with objects never seen during development, AG3S-VLA keeps 62.5% task success and 10.0% unsafe contacts (policy alone: 65.0% and 42.5%), because the only learned component is the frozen policy; unseen shapes change the surface graph, not the rule that

[TBD: Real-robot strip: the four tasks (clear, similar neighbor, carry-path obstacle, unseen objects), $\pi_{0.5}$ alone vs. AG3S-VLA, with the selected surface and the contact links overlaid on the wrist view; bar chart of task / safe success per task.]

Figure 4. **Real-world results.** The same frozen $\pi_{0.5}$ with and without AG3S-VLA on [TBD: robot], from calibrated RGB-D only; 20 trials per task and method. Unsafe contacts are scored from external sensing. Values projected.

Table 3. **Ablation on SafeLIBERO Level I (%).** One component of AG3S-VLA is replaced by its nearest alternative; everything else, including the optimizer, is unchanged. ΔSSR is the paired difference to the full method with a 95% cluster-bootstrap interval. Red values are projected.

Configuration	TSR	SSR	CAR	ΔSSR [95% CI]
AG3S-VLA (full)	56.9	**52.1**	86.3	–
Seeds: attention → nearest object	53.0	45.4	80.2	−6.7 [−9.1, −4.2]
Extent: graph → radius growth	55.8	44.9	76.5	−7.2 [−9.8, −4.5]
Extent: graph → external mask	53.6	41.8	70.9	−10.3 [−13.0, −7.4]
Identity: memory off	55.1	48.0	82.4	−4.1 [−6.2, −1.9]
Policy: target always obstacle	31.2	30.0	**92.7**	−22.1 [−25.6, −18.4]
Policy: target freed, all links/stages	**57.3**	46.1	75.8	−6.0 [−8.4, −3.5]
Policy: carried object unattached	56.4	45.2	77.0	−6.9 [−9.3, −4.4]

reads it.

4.3. Ablation Study

Tab. 3 replaces one component at a time on SafeLIBERO Level I; every variant keeps the optimizer, so the differences isolate the representation and the contact policy.

Target identity. Replacing the attention seeds by the object nearest to the hand costs 6.7 SSR points, almost entirely in the Object suite, where a distractor is closer than the target in about a third of the initial states: the rule then frees the wrong object and forbids the right one. Turning the object memory off costs 4.1 points, concentrated in episodes where the hand occludes the target at the moment of grasp and the per-request selection jumps to a neighbor.

Target extent. Growing the seeds by radius instead of along depth-continuous surfaces costs 7.2 points and lifting an external mask instead costs 10.3, in both cases through CAR rather than TSR: the freed region takes the touching obstacle with it, and the optimizer no longer sees it. This is the largest single effect among the perception components and the one an external segmenter cannot repair, since it segments in the image where touching objects share a boundary.

Contact policy. Keeping the target an obstacle in every stage gives the highest CAR (92.7%) but halves TSR, because the grasp is forbidden; freeing it for every link in every stage restores TSR but drops CAR to 75.8%, because the wrist and the carried object may now enter the target region and its surroundings. Not attaching the carried object costs 6.9 points, all in the transport phase, where the object rather than the hand hits the obstacle. The three policy terms therefore matter in the order grasp permission, attachment, link specificity, and all three are needed to reach the full method.

Table 4. **Target grounding on held-out scenes.** Soft 3D IoU, precision, and recall of the predicted observed surface against simulator ground truth, target-identification accuracy (ID), and identity switches per sequence (Sw.) in the occlusion sequences; averaged frame → episode → task. External segmenters are prompted with the noun of the instruction unless noted; the oracle receives the simulator's target label. Red values are projected.

Method	IoU	Prec.	Rec.	ID	Sw.↓
Geometry, nearest-to-hand rule	0.44	0.61	0.58	0.63	–
YOLOE [16]	0.35	0.52	0.49	0.70	5.0
Grounded-SAM-2 [12]	0.48	0.55	0.74	0.81	4.2
Grounded-SAM-2, canonical name	0.61	0.66	0.80	0.90	3.1
Oracle target (privileged)	0.80	0.93	0.85	1.00	0.0
AG3S-VLA w/o memory	0.69	0.88	0.76	0.90	2.8
AG3S-VLA (ours)	**0.72**	**0.90**	0.78	**0.96**	**0.4**

[TBD: Qualitative grounding on one held-out scene: (a) noun attention, (b) Grounded-SAM-2 mask lifted to 3D, (c) AG3S-VLA surface, (d) ground truth; rows: touching neighbor, occluded target, same scene with a changed instruction.]

Figure 5. **Grounding the observed target surface.** Attention seeds the instructed object and depth continuity bounds its surface, so the touching neighbor (blue) stays an obstacle; the lifted external mask folds it into the freed region. Changing only the instruction moves the seed, the surface, and the constraint.

4.4. Analysis

Why policy-internal grounding beats an external segmenter. Tab. 4 scores the target surface on held-out diagnostic scenes with a single target, similar objects, touching clutter, and partial occlusion. AG3S-VLA reaches an IoU of 0.72 at a precision of 0.90, against 0.48 for Grounded-SAM-2 prompted with the instruction and 0.61 when it is given the object's canonical name. The two fail differently (Fig. 5). The segmenter's recall is as high as ours, but its

precision is 0.55: when a neighbor touches the target the mask spans both, and when the robot occludes the target it follows the robot. Our precision comes from the depth-continuous graph, which stops at every visible depth break, and from the rendered robot mask. The segmenter also depends on the prompt, which costs 0.13 IoU between the canonical name and the instruction's noun, whereas the attention is conditioned on the instruction the policy actually executes; this is why KNOWS*, which selects by attention but segments externally, inherits the leakage in Tab. 1. Identity is a temporal question: the memory reduces identity switches from 2.8 to 0.4 per sequence and raises identification accuracy to 0.96, while leaving single-frame IoU almost unchanged. The remaining gap to the oracle (0.08 IoU) is observation rather than selection: unseen back faces and depth holes.

Generalization. AG3S-VLA has no learned component beyond the frozen policy, so the shift it is exposed to is perceptual. On held-out objects, layouts, and occlusions withheld from every threshold choice, its SSR drops by 1.8 points against 7.5 for KNOWS* and 9.1 for the external-segmentation baseline, whose detectors miss the unseen objects; the Level II and L2 results above and the unseen-object task on the real robot show the same pattern.

Run time. Measured from observation to returned chunk on one [TBD: GPU], AG3S-VLA answers in 290 ms (p50) and 410 ms (p95), of which the surface graph takes 95 ms and the attention read-out, taken from the forward pass that produces the actions, 6 ms; 93% of requests fit the 400 ms execution window of the deployed chunk, against 99% for AEGIS* and 88% for the external-segmentation baseline, whose segmenter dominates its cost.

References

[1] Yatharth Agarwal and Vijay Raghunathan. Multi-link safety filtering for VLA policies around moving hazards. arXiv preprint arXiv:2609.40007, 2026. 1, 2

[2] Kevin Black, Noah Brown, Danny Driess, et al. π_0: A vision-language-action flow model for general robot control. arXiv preprint arXiv:2410.24164, 2024. 2

[3] Rongxu Cui, Zongzheng Zhang, Jingrui Pang, et al. LIBERO-Safety: A comprehensive benchmark for physical and semantic safety in vision-language-action models. arXiv preprint arXiv:2606.23686, 2026. 1, 2

[4] Songqiao Hu, Zeyi Liu, Shuang Liu, Jun Cen, Zihan Meng, Shihefeng Wang, Xiang Li, and Xiao He. VLSA: Vision-language-action models with plug-and-play safety constraint layer. arXiv preprint arXiv:2512.11891, 2025. 1, 2

[5] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P. Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model. In *Conference on Robot Learning (CoRL)*, pages 2679–2713, 2024. 2

[6] Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-tuning vision-language-action models: Optimizing speed and success. In *Robotics: Science and Systems (RSS)*, 2025. 1, 2

[7] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023. 1

[8] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, et al. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, pages 38–55, 2024. 1

[9] NVIDIA. GR00T N1: An open foundation model for generalist humanoid robots. arXiv preprint arXiv:2503.14734, 2025. 2

[10] Seongbin Park, Fan Zhang, Baharan Mirzasoleiman, Shahriar Talebi, and Nader Sehatbakhsh. Your model already knows: Attention-guided safety filter for vision-language-action models. arXiv preprint arXiv:2606.09749, 2026. 1, 2

[11] Physical Intelligence, Kevin Black, Noah Brown, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. arXiv preprint arXiv:2504.16054, 2025. 1, 2

[12] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. arXiv preprint arXiv:2408.00714, 2024. 1, 3

[13] Bartolomeo Stellato, Goran Banjac, Paul Goulart, Alberto Bemporad, and Stephen Boyd. OSQP: An operator splitting solver for quadratic programs. *Mathematical Programming Computation*, 12(4):637–672, 2020. 1

[14] Balakumar Sundaralingam, Siva Kumar Sastry Hari, Adam Fishman, Caelan Garrett, Karl Van Wyk, Valts Blukis, Alexander Millane, Helen Oleynikova, Ankur Handa, Fabio Ramos, Nathan Ratliff, and Dieter Fox. cuRobo: Parallelized collision-free minimum-jerk robot motion generation. arXiv preprint arXiv:2310.17274, 2023. 1

[15] Kai Tang, Peidong Jia, Zhong Chu, Jiajun Cao, Jixian Wu, Bohua Zhang, Rui Ma, Fangyuan Zhao, Sixiang Chen, Yichen Guo, Xiaowei Chi, Chun-Kai Fan, Kevin Zhang, Jinchang Xu, Fubing Yang, Weishi Mi, Xiaozhu Ju, Jian Tang, and Shanghang Zhang. SafeDojo: Safe reinforcement learning for VLA via interactive world model. arXiv preprint arXiv:2606.20698, 2026. v2. 1, 2

[16] Ao Wang, Lihao Liu, Hui Chen, Zijia Lin, Jungong Han, and Guiguang Ding. YOLOE: Real-time seeing anything. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. 3

[17] Borong Zhang, Yuhao Zhang, Jiaming Ji, Yingshan Lei, Juntao Dai, Yuanpei Chen, and Yaodong Yang. SafeVLA: Towards safety alignment of vision-language-action model via

constrained learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. 1, 2

[18] F. Zhu, Z. Yan, Z. Hong, Q. Shou, X. Ma, and S. Guo. WMPO: World model-based policy optimization for vision-language-action models. In *International Conference on Learning Representations (ICLR)*, 2026. 1, 2